Security Project Report - A2 Trojan in the LLM era: Is LLM a plus or a minus in dealing with A2?

Amber Tsao (cctsao), Jiahao Xu (e1583425)

Introduction

Inspired by the brief survey of recent updates on how researchers tackle analog hardware trojans using LLMs, we are interested in gaining a deeper understanding of how these are done and how things might change as LLMs improve. We focus on two papers, including SPICED: Syntactical Bug and Trojan Pattern Identification in A/MS Circuits using LLM-Enhanced Detection (Chaudhuri et al., 2024) and LATENT: LLM-Augmented Trojan Insertion and Evaluation Framework for Analog Netlist Topologies (Chaudhuri et al., 2025), which respectively represent the defending and attacking LLMs of the analog hardware trojans issue.

SPICED feedback was originally designed for designers to not only detect trojans but also identify weaknesses and improve the design. However, such feedback may also help the attacker side to hide the trojan even better. According to LATENT's report, the average evasion rate can be improved to 94.4% with feedback from 23.7% for one-shot prompting. This leads us to question whether LLMs can be a solution to A2 Trojans or mainly leak additional clues to the attacker.

LATENT's percentages measure inserted components escaping localization, rather than whole circuits escaping detection. Our experiment holds the model fixed and compares feedback conditions.

Methodology

The original attack-loop experiment uses two small circuits as benchmarks, an LLM attacker inspired by LATENT's feedback loop, and a netlist-only LLM detector inspired by SPICED. In those runs, both roles used **gpt-4o-mini-2024-07-18** at temperature **0.3**. The separate detector comparison uses temperature **0** in both review conditions; the original attack-loop runs were not repeated at this temperature. These are simplified implementations, not full reproductions. We omit syntax-repair evaluation.

First, the attacker receives the clean netlist, normal and activating test conditions, and a menu of permitted changes. It returns four choices: an insertion site, a device representation, a trigger profile, and a loading strength. For example, an AND proposal can select a MOSFET branch from the output to ground, with trigger profile p2 and medium strength. Python translates these choices into actual SPICE components while preserving the original circuit.

Next, ngspice runs the modified circuit under normal inputs and an activating burst. Separately, the detector receives only the candidate netlist and its nominal function. It returns a Boolean Trojan verdict, confidence, suspicious identifiers and an explanation. The detector sees neither the clean reference nor simulation results. We count an escape only if the circuit passes both electrical checks and the detector returns is_trojan=false. Simulator results are recorded for evaluation but are not fed to either LLM.

The experiment is conducted under 3 feedback conditions to test if different levels of detail feedback affect evasion rate. “None” means that the attacker receives no feedback from the reviewer; “Binary” means that the attacker receives only the result of whether the attack is detected or not; “Full” means that suspicious nodes and explanation being found by the detector is provided back to the attacker.

Each request produces a complete next configuration. The loop does not retain a best-so-far candidate or stop after an escape; it continues for the fixed number of attempts. Thus, feedback can lead to a useful revision, an unnecessary change, or a return to an earlier design.

Methodology (continued)

Binary and full also receive their preceding candidate's choices; none has no history. Full includes verdict and confidence. Feedback goes only to the attacker. The detector uses fresh requests with fixed prompts and weights; its control-based prompt calibration occurred before the final attacks.

Each session has ten attempts: one initial proposal and nine further requests. Two sessions in each of two execution batches give **four sessions and 40 attempts per circuit/condition**. Batches use different predeclared seeds with the same model, prompts and circuits. The initial candidate and review are shared across conditions within each session. None makes fresh requests; binary and full revise from the preceding attempt.

Circuits and electrical checks

The two circuits provide readily checked analog and digital behavior while keeping simulation manageable. We verify their clean versions at 3.3 V before modification:

Clean circuit	Verification
Two-stage op-amp	Open-loop gain 95.47 dB; -3 dB bandwidth 131.90 Hz; unity-gain frequency 9.208 MHz; phase margin 50.38 degrees.
Four-input AND	16/16 logic cases pass. Rise delay 0.1476 ns; fall delay 0.1147 ns when all four inputs fall together (100 fF load, 100 ps edges).

The trigger works like a leaky bucket: input transitions add charge through a capacitor and two diodes to a 10 pF reservoir; a resistor drains it. Rapid activity can cross a threshold, enabling a branch that pulls an output or internal node toward a supply rail. These are A2-inspired simulation models, not fabricated hardware.

The op-amp permits output, gain-node or mirror-reference insertions; AND permits output or internal NAND-node insertions. Each site has three representations (switch, smooth conductance or MOSFET), three RC/threshold profiles and three strengths: **81 op-amp and 54 AND choices**. The MOSFET uses an ideal reservoir comparator. This finite menu avoids arbitrary syntax failures.

For validity, the op-amp follower must stay within 20 mV of its clean reference during normal operation and fall at least 0.3 V below it after activation. AND must preserve all 16 logic cases, stay within 0.2 V at settled outputs and lose at least 0.8 V from its activated high output. Thus a harmless modification or permanently broken circuit does not qualify.

The op-amp changes from 1.65 V +/- 50 mV at 10 kHz to a 0.6-2.7 V, 1 MHz burst at 5-45 us; its drop is measured at absolute time 50-60 us. Both amplitude and frequency change. AND receives a 1 MHz burst on a with b, c and d high. All 135 configurations were qualified before the LLM runs and each candidate was re-simulated; validity is not unrestricted LLM design ability.

Separate comparison: normal-operation evidence

We additionally review the 29 observed netlists plus two clean and two known-positive controls, deduplicated to 33 circuits. Each receives three fresh paired reviews: netlist alone versus the same netlist with normal-test summaries. Both arms use the same model snapshot as above at temperature **0**, a shared prompt and matched seeds per pair (best-effort consistency), in randomized order. This is separate from the temperature-0.3 attack experiment. Summaries contain normal I/O samples, all internal-node voltage statistics and supply current, extracted uniformly; they contain no activation tests, golden-reference differences or labels. All 198 scheduled reviews are retained without retuning. Neither the attacker nor its menu is changed, and the defender does not learn online.

Results

The six groups contain 240 analysis rows: 2 circuits x 3 conditions x 4 sessions x 10 attempts. Eight shared initial events each appear in three conditions, giving 16 reused rows and 224 actual candidate/review events. Repeated choices reduce these to 29 distinct netlists. All 224/224 candidate events pass the electrical tests. Two reviews are unusable because their responses were truncated; they remain non-successes in the per-attempt denominators, not positive detections. One subsequent binary request retained the preceding choices without a verdict.

An **escape** requires electrical validity and a usable negative Trojan verdict. The original outcome is escapes per attempt. To separate one request from a larger request budget, we also retrospectively summarize the saved first attempt and whether any of ten attempts escaped. These request-budget summaries reuse the original attack records.

Circuit	Feedback	First attempt	Any within 10	All attempts
Two-stage op-amp	none	0/4	0/4	0/40 (0.0%)
Two-stage op-amp	binary	0/4	4/4	13/40 (32.5%)
Two-stage op-amp	full	0/4	3/4	13/40 (32.5%)
Four-input AND	none	2/4	4/4	10/40 (25.0%)
Four-input AND	binary	2/4	4/4	9/40 (22.5%)
Four-input AND	full	2/4	3/4	6/40 (15.0%)

Table 1. First-attempt and any-within-ten columns count sessions (four per group); the original all-attempt column counts requests (40 per group). Initial outcomes are shared across conditions. Repeated netlists and unsuccessful reviews are retained.

Supplementary paired detector comparison

Table 2 counts positive Trojan verdicts on fixed circuits, separately from the attack escape rates above. Three fresh judgments per circuit give 198 requests, of which 195 produce usable verdicts.

Population (distinct circuits)	Netlist only: flags	Normal logs: flags
Op-amp: Observed candidates (20)	39/60	10/60
AND: Observed candidates (9)	21/27	27/27
Op-amp: Clean control (1)	0/3	0/3
AND: Clean control (1)	0/3	3/3
Op-amp: Known-positive control (1)	2/3	0/3
AND: Known-positive control (1)	3/3	3/3

Table 2. Flags over all scheduled reviews, including failures. The three truncated reviews occur in netlist-only op-amp candidates: 39 flags, 18 negative verdicts and 3 unusable replies. With logs, these become 10 flags and 50 negative verdicts. Every other cell has no unusable replies. Controls are repeated judgments of particular designs, not held-out accuracy estimates.

Normal logs do not produce a general improvement. They increase AND candidate detection from 21/27 to 27/27, but also flag its clean control 3/3 times: all 33 AND reviews in that condition are positive. Op-amp detection instead drops from 39/60 to 10/60. Among its 57 usable pairs, 33 change from positive to negative and only two change in the other direction.

Results (continued)

Consistency between execution batches

Circuit	Batch 1: none / binary / full	Batch 2: none / binary / full
Two-stage op-amp	0 / 5 / 6	0 / 8 / 7
Four-input AND	4 / 6 / 5	6 / 3 / 1

Table 3. Escape counts, each out of 20 attempts. Both batches contain two ten-attempt sessions per group; neither batch is a one-shot condition.

Full slightly exceeds binary for the op-amp in batch 1, but binary exceeds full in batch 2. AND also changes between batches. Pooling increases the number of observations; it does not remove this variation or establish a general none < binary < full ordering.

Controls in the original attack experiment

Two prompt refinements using recorded control results preceded every attack in the final experiment. Final controls reuse those benchmark designs and are **not held out**. They test repeated judgments of four particular circuits, not general detector accuracy. We set a gate of at most one clean flag and at least four known-positive flags in five reviews of each design. Batch 1 has 0/5 clean flags and 4/5 positive flags for both circuits.

Batch 2 again has 0/5 clean flags but only **3/5** positive flags for each circuit, failing the unchanged gate. After batch 1 results and batch 2 controls were known, an explicit continuation-rule **deviation** was recorded before batch 2 attacks. Prompts, seeds, menus and attempt counts were unchanged, and controls were not rerun. Batch 2 is therefore a same-settings stability assessment, not confirmation that passed the original gate.

Why repeat a request with no feedback?

Fresh model calls can select different configurations or give different reviews of the same configuration. The no-feedback condition gives this repeated sampling the same ten-request budget as the revision conditions. In fact, both AND no-feedback groups chose exactly the same output-to-ground MOSFET, trigger profile p2 and medium strength on every attempt. This design escaped 4/20 times in batch 1 and 6/20 in batch 2 without being modified.

The design is electrically effective: normal output differs by only 30.926 microvolts, while activation lowers its high output by 2.634 V. Across all conditions, this identical netlist is flagged in 42/60 actual reviews. A negative review describes the extra components as not altering the intended function, contrary to the simulated disturbance. Repeated misses here reflect inconsistent interpretation of the same circuit, not successive improvements to it.

The revision conditions also change which designs are reviewed. Binary and full each select the internal NAND node for 10/20 attempts in *each* batch. All 40 actual reviews of those internal-site candidates flag a Trojan; these are repeated events, not 40 distinct designs. Thus both spend half their attempts on consistently detected choices. Full does not use this site more often than binary, so that alone cannot explain their difference.

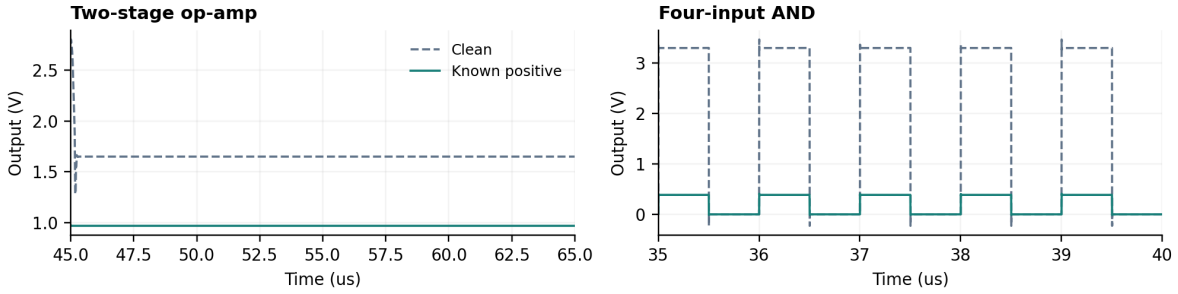


Figure 1. Clean and known-positive outputs: op-amp after the burst and AND during the burst. These activation traces establish electrical effects; neither review condition receives them.

Discussion

Feedback and search choices

In the original attack experiment, op-amp revision conditions produce more escapes than none, but full and binary tie at 13/40. AND shows the reverse per-attempt order: none 10/40, binary 9/40, full 6/40. Detailed feedback therefore has no consistent advantage. Revisions explore alternatives but can leave an intermittently missed design for a detected one. Candidate history differs between none and revision conditions; circuit complexity and menu size were not varied independently. These results do not isolate their causal effects.

What the normal logs changed

All three log-assisted reviews of the clean AND cite its internal n2 maximum of 4.35 V, above the 3.3 V supply, and a negative output value. Both features already occur in the uniformly sampled data from the unmodified clean model, whose 16 logic cases pass. They do not establish a Trojan. Conversely, some negative op-amp reviews invoke normal operation as evidence against an unauthorized modification, even though the candidate's separate activation test demonstrates a harmful effect. These are recorded rationales, not proof of the model's internal reasoning. They illustrate two risks: treating ordinary observed voltage features as malicious, and treating normal tests as proof that a conditional mechanism is harmless.

Relation to SPICED and LATENT

SPICED combines netlists, voltage/current logs, specifications and few-shot anomaly rules [2, Sec. IV and Table III]. Our supplement supplies only normal-operation summaries, including supply current rather than per-transistor currents; it does not reproduce SPICED's activation-informed diagnosis. More evidence can help only if its meaning and test coverage are interpreted correctly.

LATENT's 23.7% and 94.4% figures concern inserted components escaping localization in resulting designs [3, Table I], whereas we count whole-candidate negative verdicts over all attempts. LATENT uses broader component/node edits, simulation-informed diagnosis, reversion to a previous undetected configuration and 6-12 iterations; it studies DC-activated Trojans [3, Secs. IV-V]. We use fixed ten-attempt sessions, finite choices, no rollback and transient activation. LATENT also uses GPT-4o-mini at temperature 0.3; we did not test model progress.

Attacker or defender advantage?

The two studies show that neither richer attacker feedback nor additional normal-test evidence guarantees a better outcome. The supplement measures reviews of fixed, previously generated circuits; it does not test new adaptive attacks against the supplemented detector. Both studies keep the reviewer fixed, so neither measures online learning. Without a non-LLM comparison, we cannot quantify which side benefits more overall. Four attack sessions per group, reused calibration controls, the original repeat gate deviation and simulations at a single supply and nominal model settings limit generalization. The control designs are not held out, and repeated reviews do not increase their number.

References

[1] K. Yang et al. "A2: Analog Malicious Hardware." IEEE S&P, 2016, pp. 18-37. doi:10.1109/SP.2016.10.

[2] J. Chaudhuri et al. "SPICED: Syntactical Bug and Trojan Pattern Identification in A/MS Circuits using LLM-Enhanced Detection." 2024. arXiv:2408.16018v1.

[3] J. Chaudhuri, A. Chaudhuri and K. Chakrabarty. "LATENT: LLM-Augmented Trojan Insertion and Evaluation Framework for Analog Netlist Topologies." 2025. arXiv:2505.06364v1.

Code and evidence: github.com/MikeJiahaoXu/a2-llm-trojan-study. AI tools assisted implementation and writing; measurements come from saved responses and simulations.